(13)

# ASSESSMENT TOOL - I

## ANALYTICAL PROBLEM SOLVING (1)

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Q1) Convert the expression ((a+b)+c*(d+e)+f)*(g+h) to
1) Prefix expression  2) Postfix expression

Sol: Given expression is:
((a+b)+c*(d+e)+f)*(g+h)

1. Prefix expression

| character | stack | output |
|---|---|---|
| ) | ) | - |
| h | ) | h |
| + | )+ | hg |
| ( | Empty | hg+ |
| * | * | hg+ |
| ) | *) | hg+ |
| f | *) | hg+f |
| + | *)+ | hg+f |
| ) | *)+) | hg+f |
| e | *)+) | hg+fe |
| + | *)+)+ | hg+fe |
| d | *)+)+ | hg+fed |
| ( | *)+ | hg+fed+ |
| * | )+ | hg+fed+ |
| c | )+ | hg+fed+c |
| + | *)+ | hg+fed+c*+ |
| ) | *)+) | hg+fed+c*+ |
| b | *)+) | hg+fed+c*+b |

| | | |
|---|---|---|
| + | *)+)+ | hg+fed+c*+b |
| a | *)+)+ | hg+fed+c*+ba |
| ( | *)+) | hg+fed+c*+ba+ |
| ( | * | hg+fed+c*+ba++ |
| End | Empty | *+++ab*c+def+gh |

2) Postfix Expression

| character | stack | output |
|---|---|---|
| ( | ( | - |
| ( | (( | |
| a | (( | a |
| + | ((+ | a |
| b | ((+ | ab |
| ) | ( | ab+ |
| + | (+ | ab+ |
| c | (+ | ab+c |
| * | (+* | ab+c |
| ( | (+*( | ab+c |
| d | (+*( | ab+cd |
| + | (+*(+ | ab+cd |
| e | (+*(+ | ab+cde |
| ) | (+* | ab+cde+ |
| + | (+ | ab+cde+*+ |
| f | (+ | ab+cde+*+f |
| ) | Empty | ab+cde+*+f+ |
| * | * | ab+cde+*+f+ |
| ( | *( | ab+cde+*+f+ |
| g | *( | ab+cde+*+f+g |
| + | *(+ | ab+cde+*+f+g |
| h | *(+ | ab+cde+*+f+gh |
| ) | * | ab+cde+*+f+gh+ |
| End | Empty | ab+cde+*+f+gh+ |

Two students obtain different postfix expressions for same infix expression. Infix: (A+B)*(C-D)/E

Student A: AB+CD-*E/

Student B: AB+C*D-E/

Determine which postfix expression is correct and justify your answer by tracing the conversion process.

Given that:

Infix Expression is: (A+B)*(C-D)/E

Table for Postfix:

| Character | Stack | Prefix |
|---|---|---|
| ( | ( | - |
| A | ( | A |
| + | (+ | A |
| B | (+ | AB |
| ) | Empty | AB+ |
| * | * | AB+ |
| ( | *( | AB+ |
| C | *( | AB+C |
| - | *(- | AB+C |
| D | *(- | AB+CD |
| ) | * | AB+CD- |

| | | |
|---|---|---|
| / | / | AB+CD-* |
| E | / | AB+CD-*E |
| End | Empty | *AB+CD-E/ |

Final postfix:

$\boxed{AB+CD-*E/}$

∴ Student A expression is correct.

Reason:

The Infix expression is (A+B)*(C-D)/E. First, (A+B) and (C-D) must be evaluated. then multiplication * is performed, followed by divison /. Student A follows correctly, while student B performs C* before completing (C-D), so it does not represent the given infix expression.